

Middle-Mile Line-Haul Network Optimization

Problem Statement

Planning horizon: 31 days (01-07-2026 to 31-07-2026)

1 Context

A B2B line-haul network moves shipments between sort centres on hired vehicles. Today most freight moves point-to-point on the origin-destination (OD) lane it was booked on. Direct movement guarantees the shortest distance and zero intermediate handling, but it forces vehicles to depart under-filled on thin lanes and pays the fixed cost of a trip many times over.

The alternative is consolidation: route freight from several thin lanes through a common intermediate node so that fewer, fuller vehicles run the long legs. Consolidation buys vehicle utilisation but pays for it three ways - extra distance (the detour through the intermediate node), extra handling cost (unloading and reloading at that node), and extra transit time.

The problem is to find the routing and fleet plan that achieves minimum total cost, subject to the physical and service constraints set out below.

2 Objective

Minimise the total cost of running the network over the planning horizon. Total cost is the sum of four components:

1. **Fixed cost** - the fixed cost of the vehicle type used, charged once for every trip dispatched.
2. **Per-km cost** - the per-km cost of the vehicle type used, multiplied by the distance that trip travels.
3. **Hop cost** - charged whenever freight is unloaded and re-handled at an intermediate node. It is the cost per kg at that node multiplied by the weight handled there.
4. **Touch cost** - charged at half the hop rate when a vehicle stops at a node but the freight in question is not unloaded (a touching connection). It is half the cost per kg at that node, multiplied by the weight carried through.

Where a trip runs one-way, its fixed and per-km cost is adjusted by the one-way factor for that lane, as described in Section 3.

Component	Source	Basis
Fixed cost	vehicle_data_full.csv, column Fixed cost	per dispatched trip, by vehicle type
Per-km cost	vehicle_data_full.csv, column Per KM cost (Fixed)	per km travelled, by vehicle type
Hop cost	B2B Loader CPK...csv, column CPK	per kg handled, specific to the node where handling occurs
Touch cost	same, at 50 %	per kg, at the node touched
One-way factor	one_way_lane.xlsx, col- umn %age	share of round-trip cost charged, by lane

A shipment moving from A to B direct incurs no hop cost. The same shipment moving from A to B via X, with an unload and reload at X, incurs the full hop cost at X on its own weight. If

the vehicle merely calls at X to drop other freight and this shipment stays loaded, the shipment incurs half the hop cost at X.

3 One-Way Lane Pricing

Vehicles are costed on a round-trip basis. A one-way movement therefore does not cost half of a round trip by default, because the empty return is still paid for. The one-way lane table defines the exceptions.

For a lane listed in `one_way_lane.xlsx`, a one-way trip on that lane costs the given percentage of what the round trip on the same lane would cost. For every other lane, a one-way trip costs the same as its round trip. The factor applies to the whole trip cost, fixed plus per-km.

4 Data

4.1 Demand Data (`monthly_clustered_loads_combined.csv`)

Column	Meaning
<code>date</code>	dispatch date, DD-MM-YYYY, 31 distinct days
<code>source_node, dest_node</code>	origin and destination sort centre
<code>total_Phy_wt</code>	physical weight, kg
<code>total_Vol_wt</code>	volumetric weight, kg
<code>total_shipments</code>	number of shipments in the cluster
<code>detour_factor</code>	maximum permitted routing detour for that OD pair (see C8)

552 distinct OD pairs across 24 demand nodes.

4.2 Node Data (`node_processing_120.csv`)

Columns: `node`, `processing_capacity_kgs`, `consider`, `processing_time_hrs`, `consolidation`.

All 24 nodes are flagged `consider = Yes` and `consolidation = Y`, so every node is a candidate consolidation point. Processing time is 2 hrs everywhere.

`processing_capacity_kgs` is the node's *total* daily throughput limit, covering incoming and outgoing load together.

4.3 Distance Data (`one_to_all_distance_matrix_full.csv`)

Columns: `origin`, `destination`, `key` (formatted "origin - destination"), `time_hrs` and `distance_km`. A complete 24 by 24 matrix over the 24 demand nodes, 576 rows.

4.4 Fleet Data (`vehicle_data_full.csv`)

Vehicle Type	Phy Cap (kg)	Vol Cap (kg)	Fixed cost	Per-km cost	Round-trip km limit
32 FT (Multi Axle)	18,000	12,920	3,364	33.17	unbounded
32 FT Single Axle (9-MT)	9,500	12,920	2,515	30.80	unbounded
24 FT	6,500	7,680	1,711	26.92	2,000
22 FT	6,500	7,013	1,604	25.55	2,000
20 FT	6,000	6,773	1,489	24.41	2,000
17 FT	5,000	4,463	1,145	18.70	1,000
14 FT	3,500	3,528	977	17.03	1,000
10 FT	2,500	2,205	769	15.91	800
Mahindra Pickup	1,200	1,472	563	13.16	600

The two 32 FT types carry a nominal 100,000 km limit, i.e. effectively unbounded. Every type allows a maximum of 4 intermediate stops. Vehicle availability is unlimited by type; fleet size is a decision (see C7).

4.5 Hop Cost Data (B2B Loader CPK Working March-26.csv)

Columns: Period, Location, CPK (cost per kg). Covers all 24 demand nodes.

4.6 Lane Cost Factors (one_way_lane.xlsx)

Lane-level %age values giving the share of round-trip cost charged for a one-way trip on that lane, as set out in Section 3, plus the roster of OD pairs designated one-way.

5 Hard Constraints

C1 - Vehicle capacity. On every leg of every trip the freight on board must satisfy both limits simultaneously: total physical weight within `Phy Cap(Kgs)` and total volumetric weight within `Vol Cap(Kgs)` of the assigned vehicle type.

C2 - Node processing capacity. At every node on every day, total load handled - incoming plus outgoing, including freight passing through on a consolidation hop - must not exceed the node's `processing_capacity_kgs` in `node_processing_120.csv`. That figure is already the combined in-plus-out limit.

C3 - Maximum round-trip distance. A vehicle's total round-trip distance must not exceed the `Round Trip KM Range` for its type. The two 32 FT types are effectively unbounded; smaller types are restricted to 600–2,000 km round trip, which confines them to short-haul and feeder roles.

C4 - Maximum intermediate stops. No trip may make more than 4 intermediate stops (`max_intermediate_stops`), excluding origin and final destination.

C5 - Directional routing. A vehicle's stop sequence must progress consistently towards its destination. Back-tracking and zig-zag routes are prohibited: a route of the form DEL - BLR - HYD - MAA - DEL, which overshoots and doubles back, is invalid even if it satisfies the distance cap. In practice, each successive stop must not increase the remaining distance to the trip's terminal node, or must satisfy an equivalent measure of forward progress agreed with the business.

C6 - Path stability over the month. An OD pair's volume may be split across multiple paths, but each split portion must follow the same path every day of the month. Path assignment is decided once for the horizon; daily re-routing of the same freight stream is not permitted. This keeps the operating plan stable and auditable.

C7 - Constant daily fleet. The number of vehicles of each type deployed must be identical on every day of the month.

C8 - Detour factor. For each OD pair, the actual routed distance must not exceed the detour factor for that pair multiplied by the direct distance between its origin and destination. Values in the demand file are 1.2 (136 OD pairs), 1.5 (202), 3.0 (212), 4.0 (1) and 5.0 (1), constant per OD pair across the month. Tight factors sit on the high-volume trunk lanes; loose factors on thin lanes where consolidation is the only economic option.

C9 - Unassigned spillover within 12 %. Because the fleet is fixed and identical on every day (C7) while daily demand varies, some load will inevitably go unassigned on the heavier days. This spillover is permitted, but it must stay within 12 % of the total load. The limit applies separately to physical weight and to volumetric weight, so a plan that spills within the limit on one measure but breaches it on the other is not acceptable. Report the spilled load explicitly rather than dropping it silently from the plan.

6 Soft Constraints (Good to have)

S1 - Weighted hops per kg within 0.2. Taking every shipment's hop count weighted by its weight, divided by total weight, the network average must stay within 0.2. In effect, at most 20 % of tonnage may take a single intermediate hop; equivalently, at least 80 % of freight by weight must move direct.

S2 - Weighted average distance per kg within 1,300 km. Taking every shipment's routed distance weighted by its weight, divided by total weight, the network average must stay within 1,300 km.

S3 - Single path between any hub pair. For every ordered pair of nodes there may be at most one shipment-transfer path, and every order whose route touches both of those nodes must traverse exactly that path between them. If SC_1 to SC_4 is routed as SC_1 - SC_2 - SC_4, then any order using SC_1 and SC_4 as hops must follow SC_1 - SC_2 - SC_4 between those two points. The routing plan collapses to a single next-hop table, which is what makes it operable on the ground.

Report S1, S2 and S3 compliance for every candidate solution. Where a target cannot be met, quantify the trade-off: the cost of meeting it against the cost of the best unconstrained solution.

7 Deliverables

The solution must produce five reports. Together they should let a reader trace any single order from its origin to its destination, see which vehicles carried it and how the cost was built up, and check every constraint in Sections 5 and 6 without re-running the model.

7.1 Order-level routing report

One entry per order. Each entry should identify the order and its origin and destination, record its physical and volumetric weight, and state whether the order was served or left unserved.

It should then describe how the order actually moved: the full route in sequence from origin to destination, how many legs that route contains, how many transshipment stops it passes through, and which nodes those are. Alongside the route, record the direct distance and direct travel time between origin and destination, the distance and travel time actually incurred, and the difference between the two in both absolute terms and as a ratio against the direct distance. Include the processing time accumulated at intermediate nodes, the total elapsed time once processing is added, and the handling cost the order incurred at its hops.

Each entry should also name every vehicle that carried the order, in leg order, so that the chain of custody is readable end to end. It should carry the cost apportioned to that order, and a plain pass or fail marker against the stop limit and the detour cap, so that non-compliant orders can be isolated without recomputing anything. Where the order was moved as part of a return movement, record that link and the load carried back.

7.2 Hop view of the order-level report

The same order-level view, extended to make the hop metric auditable. For each order, record the number of transfer hops it incurred and the path expressed as the sequence of hub-to-hub transfers rather than as a vehicle itinerary.

7.3 Order-to-vehicle allocation report

One entry per combination of leg, order and vehicle. This is the record that makes load splitting visible: where an order is carried by more than one vehicle on the same leg, or moves across different vehicles on successive legs, each portion appears as its own entry.

Each entry should name the leg by its start and end node, identify the order and the specific

vehicle, and state the physical and volumetric weight assigned to that vehicle together with the share of the order it represents. Where a portion needs attention, for instance an unusually small split or a load that only just fits, flag it in the entry.

7.4 Vehicle route report

One entry per vehicle deployed. Each entry should identify the vehicle and its type, name the node it starts from, and set out its complete round-trip route in stop sequence, returning to its start.

Against that route, record the number of legs run and the maximum permitted, the outbound distance, the return distance and the round-trip total, and the range limit for that vehicle type. Both the leg count and the distance should carry a plain pass or fail marker against their limits. Record the total travel time of the trip.

On the load side, record how many orders the vehicle consolidated and list them, state the physical and volumetric load carried outbound, and set both against the vehicle's rated capacities to give utilisation on each dimension separately, plus a single effective utilisation figure. Record how the return leg was handled, whether it was a closed circuit, a loaded return, or run empty, and identify the orders carried back where there were any.

On the cost side, break the trip cost into its fixed and per-km components and give the total, with the one-way factor already applied where the trip ran one-way. Report the full line-haul cost of the trip alongside it.

Finally, each entry should flag whether the vehicle's route matches the path the order was assigned in the routing report, and whether it respects the single-path rule for the hub pairs it touches. A mismatch on either should be named explicitly rather than left for the reader to detect.

7.5 Network summary

A single consolidated summary of the whole plan, grouped so that each part of the problem can be read on its own.

Cover the order position: how many orders were in the dataset, how many were served, how many were left unserved, and the resulting service rate. Report the spilled load alongside it, on physical weight and volumetric weight separately, as a share of total load against the 12 % limit. Cover the fleet: the total vehicle count and its breakdown by vehicle type. Cover distance: the total distance run across the network, split into loaded and empty running, with the number of empty legs called out. Cover cost: fixed and per-km cost and their total, the full line-haul cost, and the same figures normalised per order served and per kilometre run.

Cover utilisation and consolidation: average and peak vehicle utilisation, how many orders moved direct against how many were consolidated, and the resulting consolidation rate. Cover compliance: the count and percentage of orders meeting the stop limit and the detour cap, the network figures for weighted hops per kg and weighted average distance per kg against their targets, and whether the single-path rule holds across all hub pairs. Cover the nodes: average and peak node utilisation against **processing_capacity_kgs**, and which node is the busiest.